

Knee laxity measurements after anterior cruciate ligament reconstruction, using either bone–patellar–tendon–bone or hamstring tendon autografts, with special emphasis on comparison over time

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Received: 9 February 2009 / Accepted: 8 June 2009 / Published online: 3 July 2009
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Abstract The aims of the study were to analyse the change in knee laxity over time after anterior cruciate ligament (ACL) reconstruction, using either bone–patellar–tendon–bone (BPTB) or hamstring (HS) tendon autografts, and to compare the knee laxity measurements between the study groups both pre-operatively and on multiple follow-up occasions. Another aim was to compare the radiographic findings in terms of degenerative changes between the study groups. A randomised series of 71 patients, who underwent ACL reconstruction using BPTB or HS tendon autografts and interference screw fixation, were included in the study. Of these patients, 47/71 (66%) attended a clinical examination, including laxity measurements using the KT-1000 arthrometer, pre-operatively and on four post-operative occasions; 6 months, 1 year, 2 years and 7 years after the reconstruction. The BPTB group consisted of 22 patients, while there were 25 patients in the HS group. There were no significant differences in the mean side-to-side knee laxity between the BPTB and the HS group pre-operatively or at the follow-up examinations. There was a tendency towards a reduction in side-to-side knee laxity over time in both groups, measured with the KT-1000 arthrometer. The decrease was significant when analysing

the injured and uninjured knee separately (injured side $p < 0.001$ (BPTB) and $p = 0.005$ (HS), uninjured side $p = 0.008$ and $p = 0.042$, respectively). Forty-four patients (BPTB 21, ST 23) underwent a radiographic assessment at the 7-year follow-up, which revealed no significant differences between the study groups in terms of osteoarthritic findings classified according to the Fairbank and Ahlbäck rating systems. In overall terms, osteoarthritis was identified in 16% (BPTB 19%; ST 13%; n.s.) according to the Ahlbäck rating system and 68% (BPTB 67%; ST 70%; n.s.) according to the Fairbank rating system. There were no significant differences in knee laxity measurements between the two study groups pre-operatively or at 7 years. A decrease in knee laxity over time was seen in both groups. There were no significant differences between the BPTB and ST groups in terms of osteoarthritic findings at 7 years.

Keywords ACL reconstruction · Knee laxity · KT 1000 arthrometer · Bone–patellar–tendon–bone autograft · Hamstring tendon autograft

Introduction

During the past 20 years hamstring tendon autografts (HS), e.g. semitendinosus or semitendinosus and gracilis, have been used more and more frequently and they are currently probably used as frequently as the bone–patellar–tendon–bone (BPTB) autograft, when reconstructing an anterior cruciate ligament (ACL) injury. The advantages and disadvantages of these two types of autograft are currently an ongoing topic for discussion and evaluation. Several meta-analyses have been conducted in order to spotlight the

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differences and, if possible, recommend either type of graft [3, 13, 14, 16, 45]. The arguments in favour of the BPTB autograft are fewer graft failures and less knee laxity. Correspondingly, the arguments in favour of the HS autograft are less anterior knee pain and less extension deficit. However, we have found no study in the literature analysing knee laxity over time for the HS-autograft separately. The selection of graft is, however, often a result of the surgeon's experience and preference and also what appears to be most appropriate for the individual patient [39].

ACL injuries are common among young athletes and the main aim for reconstruction is to regain knee stability and the ability to return to the pre-injury sport activity level in the short term. However, long-term follow-ups are also of major importance for young patients. The functional, objective and subjective outcomes have been evaluated 5–10 years after ACL reconstruction [2, 30–32, 34]. In the long-term after ACL reconstruction, osteoarthritis (OA) has been reported in a large percentage of patients, especially after using the BPTB autograft [20, 25, 33, 40, 44]. The question of whether laxity in the ACL-reconstructed knee increases over time has also been discussed [4, 8].

The aims of the present study were to analyse the changes in knee laxity over time, after the index ACL reconstruction using either a BPTB or HS autograft. Furthermore, to compare the knee laxity measurements between the two study groups both pre-operatively and at follow-up.

The hypothesis was that there are no significant changes in knee laxity over time during the post-operative follow-up period in either the BPTB or HS groups. Moreover, the two groups are comparable in terms of knee laxity measurements during the follow-up period. Another hypothesis was that there are no significant differences between the study groups in terms of osteoarthritic findings.

Patients and methods

Patients

Between April 1995 and May 1998, 71 patients who underwent an ACL reconstruction using either an ipsilateral BPTB autograft or an ipsilateral triple or quadruple ST autograft were included in a randomised series [9]. The inclusion criterion was unilateral ACL injury, verified by clinical examination or arthroscopy. The exclusion criteria were posterior cruciate ligament injury, more than +1 medial or lateral collateral ligament injury, re-injury during the follow-up period and radiographically visible OA at the time of inclusion. Of the patients, 47/71 (66%) attended a clinical examination pre-operatively and on all the four

scheduled post-operative occasions, 6 months (Fp I), 1 year (Fp II), 2 years (Fp III) and 7 years (Fp IV) post-operatively. The BPTB group consisted of 22 patients, while there were 25 patients in the ST group. The groups were comparable in terms of pre-operative demographics (Table 1). The cause of injury is presented in Table 2.

Surgical technique

One senior surgeon performed all the reconstructions. Associated intra-articular injuries, such as meniscal ruptures and chondral lesions, were addressed at the time of the index operation.

BPTB technique

The arthroscopic transtibial technique and interference screw fixation were used during the index procedure [23]. The mid-third of the patellar tendon was harvested through two 25-mm long vertical incisions, one over the apex of the patella and the other just above the tibial tubercle. The graft was tunnelled subcutaneously under the paratenon with the aim of protecting the infrapatellar nerve and its branches

Table 1 Descriptive data

	Group BPTB	Group ST	Significance
Number of patients	22	25	n.s. (1.0)
Age (median, years)	26 (14–48)	29 (15–40)	n.s. (0.86)
Gender female/male	8/14	7/18	n.s. (0.54)
Injured side left/right	13/9	14/11	n.s. (0.83)
Time between the injury and index operation (months, median and range)	11 (2–252)	17 (3–240)	n.s. (0.39)
Follow-up I (months, median and range)	6 (5–8)	6 (5–7)	n.s. (0.20)
Follow-up II (months, median and range)	12 (11–28)	12 (11–24)	n.s. (0.55)
Follow-up III (months, median and range)	25 (23–30)	24 (23–30)	n.s. (0.93)
Follow-up IV (months, median and range)	89 (77–110)	86 (69–109)	n.s. (0.12)

Table 2 Cause of injury

	Group BPTB <i>n</i> = 22	Group ST <i>n</i> = 25	Significance
Contact sport	18	16	n.s. (0.49)
No contact sport	3	5	
ADL		1	
Other	1	3	

and leaving the major part of the paratenon intact, as described previously by Kartus et al. [21]. The proximal bone block was sized to 9 mm and the distal bone block to 10 mm. The bone tunnels were prepared in a standard transtibial fashion. The femoral tunnel was drilled transtibially and placed at approximately the 10.30 position in the right knee and the tibial tunnel was placed just anterior to the normal posterior cruciate ligament. A 7 mm and a 9 mm Acufex® (Acufex, Microsurgical Inc., Mansfield, MA, USA) “silk” interference screw were used on the femoral and tibial sides, respectively.

ST technique

The graft was harvested through an approximately 3-cm long incision over the pes anserinus. The tendons were palpated and the sartorius fascia was incised parallel to the fibres of the fascia, just above the thicker and more distally inserted ST tendon. After the vinculae had been cut under visual control, the ST tendon was harvested with a semi-blunt, semi-circular open tendon stripper (Acufex, Microsurgical Inc., Mansfield, MA, USA). Depending on its length, the ST tendon was prepared as a triple or quadruple graft. The femoral tunnel was drilled transtibially, while the tibial tunnel was drilled in standard fashion. Both the femoral and the tibial tunnels were placed at the same locations as in the BPTB group. A 7 mm soft-threaded RCI® (Smith and Nephew, Inc, Andover, MA 01810, USA) interference screw was used on both the femoral and tibial sides [6].

Standard radiography

Radiographic assessments were performed at the 7-year follow-up for 21 patients in the BPTB group and 23 patients in the ST group. Standard weight-bearing radiographic examinations using the anterior–posterior (AP) and lateral views, with 30° of knee flexion, were classified according to the Ahlbäck [1] and the Fairbank rating systems [10]. An independent experienced radiologist, blinded to the type of graft, evaluated and classified the radiographs.

Clinical assessments

One physiotherapist, who was not involved in the rehabilitation, performed all the pre- and post-operative follow-ups, according to a standard protocol. The physiotherapist was blinded to the aim of the study at the time of the examination but not to the type of surgical technique that was used.

At the pre-operative examination and the examination at all follow-ups, both groups underwent multiple subjective

and objective evaluation tests commonly used for testing ACL-injured knees, such as range of motion (ROM) [5], one-leg hop-test [42], manual Lachman’s test, Lysholm knee scoring scale [18, 41] and Tegner activity level evaluation [42]. The ROM is presented dichotomously as having or not having an extension or flexion deficit. The manual Lachman’s test was estimated by the examiner as the amount of anterior drawer movement with the knee in 15–20° of flexion. It was graded as 0, +(<5 mm), ++ (5–10 mm) or +++(>10 mm), compared with the uninjured contralateral knee [43]. For these variables, only the results from pre-operative assessments and the 7-year follow-up are presented in the study.

The knee laxity measurements using the KT-1000 arthrometer are presented for all follow-up occasions.

The instrumented KT-1000 arthrometer examination (MEDmetric® Corp, San Diego, Ca) was performed with the patient in the supine position [7]. Both legs were placed on a thigh support with the knees in 30° of flexion. A foot rest and a strap around the thighs kept the legs in a neutral position [12, 28]. The arms were placed along the sides of the body and the patient was asked to relax. The instrument was calibrated to zero before each displacement test. The anterior displacement of the tibia in relation to the femur was registered at 89N. The readings of the needle position were only accepted if the needle returned to 0 ± 0.5 mm, when the tension in the handle was released. The uninjured leg was always tested first. At least three measurements of each knee were made and the average value was registered.

Rehabilitation

All the patients were rehabilitated according to the same guidelines by their local physiotherapists, permitting immediate full weight-bearing and full ROM including full hyperextension [38]. However, no external load in open kinetic chain exercises apart from the weight of the operated leg was used during the first six post-operative weeks from 30° to full (hyper) extension [17]. No rehabilitation brace was used [22]. Closed-chain exercises were started immediately post-operatively. Running was permitted at 3 months and contact sports at 6 months at the earliest, provided that the patient had regained full functional stability in terms of strength, co-ordination and balance as compared with the contralateral leg.

Statistical methods

Mean (SD) and median (range) values are presented when applicable. For comparisons of dichotomous variables between the groups, Fisher’s exact test and the chi-square test were used. In terms of both continuous and non-continuous variables, the Mann–Whitney *U* test was used.

Friedman's and Wilcoxon's signed rank test were used for comparisons of the pre-operative and post-operative data within the study groups. A p value of ≤ 0.05 was considered statistically significant.

Ethics

The Human Ethics Committee at the Medical Faculty at Gothenburg University approved the study.

Results

There were no significant differences between the study groups in terms of cause of injury (Table 2). Both groups improved significantly between the pre-operative assessment and the long-term follow-up (Fp IV) in terms of the Tegner activity level (BPTB group $p = 0.003$ and ST group $p = 0.007$) and the Lysholm knee scoring scale (BPTB group $p = 0.008$ and ST group $p = 0.005$) (Table 3). At Fp IV, 14% in the BPTB group and 12% in the ST group displayed an extension deficit (n.s.); the corresponding values for flexion deficit were 27 and 48%, respectively (n.s.) (Table 3).

The manual Lachman's test improved significantly between the pre-operative values and Fp IV in both groups

(BPTB group and ST group $p < 0.001$, respectively) (Table 3).

The absolute values for the anterior displacements of the injured and uninjured knees, as well as the side-to-side differences, are presented in Table 4. There were no significant differences in knee laxity as measured with the KT-1000 arthrometer between the BPTB group and ST group pre-operatively or at the follow-ups.

There was a tendency towards reduced knee laxity measured with the KT-1000 arthrometer over time in both groups in terms of the side-to-side difference (Fig. 1).

This decrease was significant when analysing the injured and uninjured knees separately (injured side $p < 0.001$ and $p = 0.005$ BPTB and ST groups respectively; uninjured side $p = 0.008$ and $p = 0.042$ respectively) (Figs. 2, 3).

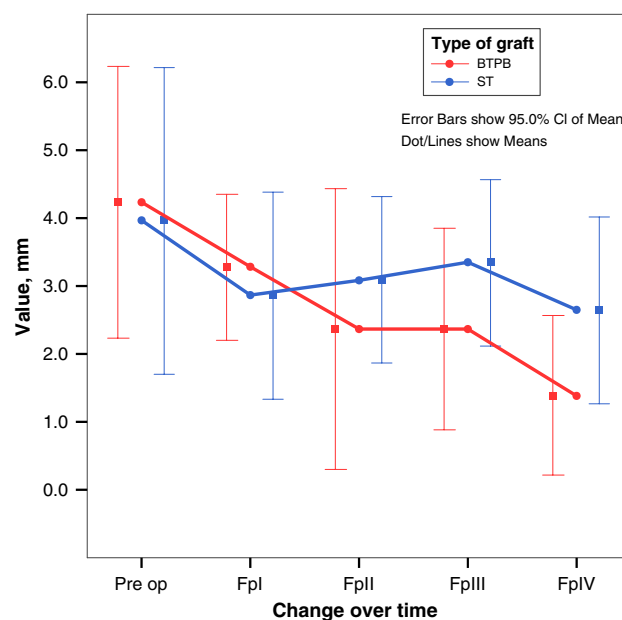
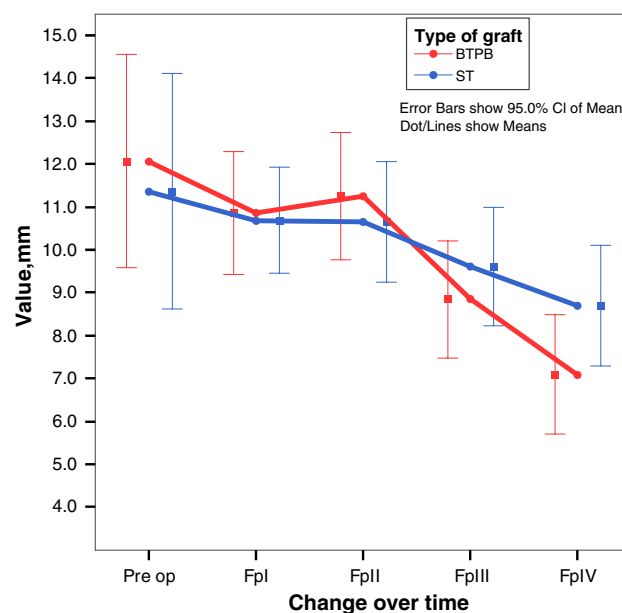
Forty-four patients (ST 23, BPTB 21, missing 3) underwent the radiographic assessment, which revealed no significant differences between the study groups at Fp IV in terms of OA findings classified according to the Fairbank and Ahlbäck rating systems (Table 5). In overall terms, OA was identified in 16% (BPTB 19%; ST 13%; n.s.) according to the Ahlbäck rating system and 68% (BPTB 67%; ST 70%; n.s.) according to the Fairbank rating system. No significant differences in the prevalence of patellofemoral osteophytes (BPTB 38%; ST 44%, in total 41%) were found between the study groups.

Table 3 Clinical assessments pre-operative and at follow-up IV

	Group BPTB $n = 22$		Group ST $n = 25$		Significance BPTB vs. ST	
	Pre-op	Follow-up IV	Pre-op	Follow-up IV	Pre-op	Follow-up IV
Manual Lachman test missing value		1 (4%)		2 (8%)	$p = 0.009$	n.s. (0.42)
0		7 (32%)		6 (24%)		
+1	1 (4%)	14 (64%)	6 (24%)	15 (60%)		
+2	5 (23%)		10 (40%)	2 (8%)		
+3	16 (73%)		9 (36%)			
Pre-op vs Fp IV		$p < 0.001$		$p < 0.001$		
Tegner activity scale (median)	3 (2–8)	5 (2–7)	4 (2–9)	5 (0–9)	n.s. (0.74)	n.s. (0.52)
Pre-op versus Fp IV		$p = 0.003$		$p = 0.007$		
Lysholm knee scoring scale (median, points)	70.5 (45–95)	81 (49–100)	67 (21–91)	89 (15–100)	n.s. (0.75)	n.s. (0.78)
Missing value				4		
Pre-op vs Fp IV		$p = 0.008$		$p = 0.005$		
One-leg hop (median, %)	67.5 (10–111)	84.5 (0–119)	86 (0–105)	94.5 (60–119)	$p = 0.01$	$p = 0.006$
Missing value		1				
Pre-op versus Fp IV		n.s. (0.12)		$p = 0.001$		
Extension deficit	5 (23%)	3 (14%)	7 (28%)	3 (12%)	n.s. (0.68)	n.s. (1.0)
Pre-op versus Fp IV		n.s. (1.0)		n.s. (0.53)		
Flexion deficit	9 (41%)	6 (27%)	5 (20%)	12 (48%)	n.s. (0.12)	n.s. (0.15)
Pre-op versus Fp IV		n.s. (0.66)		n.s. (0.65)		

Table 4 KT-1000 arthrometer measurements

	Group BPTB <i>n</i> = 22	Group ST <i>n</i> = 25	Significance
Pre-operative			
Side-to-side			
Mean (SD)	4.2 (4.5)	3.9 (5.5)	n.s. (0.47)
Median (range)	4.0 (−3.0–17.0)	2.0 (−2.0–24.0)	
Injured side			
Mean (SD)	12.1 (5.6)	11.4 (6.7)	n.s. (0.33)
Median (range)	9.75 (8.0–29.0)	9.5 (5.0–30.0)	
Uninjured side			
Mean (SD)	7.8 (3.6)	7.4 (3.8)	n.s. (0.82)
Median (range)	6.75 (4.0–17.0)	7.0 (1.5–22.0)	
Follow-up I			
Side-to-side			
Mean (SD)	3.2 (2.4)	2.9 (3.7)	n.s. (0.94)
Median (range)	3.0 (−2.0–7.0)	3.0 (−5.5–8.5)	
Injured side			
Mean (SD)	10.9 (3.2)	10.7 (3.0)	n.s. (0.86)
Median (range)	11.0 (5.5–18.0)	11.0 (4.5–16.0)	
Uninjured side			
Mean (SD)	7.6 (3.0)	7.8 (2.9)	n.s. (0.77)
Median (range)	8.0 (3.5–15.0)	8.0 (3.0–15.0)	
Follow-up II			
Side-to-side			
Mean (SD)	2.4 (4.7)	3.1 (2.8)	n.s. (0.57)
Median (range)	3.0 (−10–13.5)	4.0 (−2.0–8.5)	
Injured side			
Mean (SD)	11.3 (3.3)	10.7 (3.2)	n.s. (0.78)
Median (range)	10.0 (7.0–19.0)	11. (4.0–17.0)	
Uninjured side			
Mean (SD)	8.8 (5.0)	7.6 (3.8)	n.s. (0.39)
Median (range)	7.0 (4.0–23.0)	7.0 (3.0–16.0)	
Follow-up III			
Side-to-side			
Mean (SD)	2.4 (3.6)	3.3 (2.9)	n.s. (0.25)
Median (range)	1.5 (−4.0–11.5)	3.0 (−1.0–10.5)	
Injured side			
Mean (SD)	8.8 (4.1)	9.6 (3.3)	n.s. (0.22)
Median (range)	8.0 (4.0–16.5)	10.0 (2.5–17.0)	
Uninjured side			
Mean (SD)	6.5 (2.6)	6.3 (2.5)	n.s. (0.87)
Median (range)	6.0 (3.0–13.0)	6.0 (2.0–11.0)	
Follow-up IV			
Side-to-side			
Mean (SD)	1.4 (2.6)	2.6 (3.3)	n.s. (0.17)
Median (range)	1.0 (−3.0–5.0)	3.5 (−4.0–9.0)	
Injured side			
Mean (SD)	7.1 (3.1)	8.7 (3.4)	n.s. (0.07)
Median (range)	7.0 (2.0–16.0)	9.0 (3.0–16.0)	
Uninjured side			
Mean (SD)	5.7 (2.4)	6.1 (3.1)	n.s. (0.89)
Median (range)	6.0 (2.0–12.0)	5.0 (2.0–15.0)	

**Fig. 1** Side-to-side difference (*n* = 47)**Fig. 2** Absolute values involved side (*n* = 47)

Discussion

The principal finding in this study was that there was a reduction in knee laxity over time after an ACL reconstruction using either a BPTB or ST autograft up to 7 years after the index operation. This decrease was statistically significant, when analysing the absolute values for the injured and uninjured sides separately. However, the decrease did not reach statistical significance for the difference in side-to-side laxity.

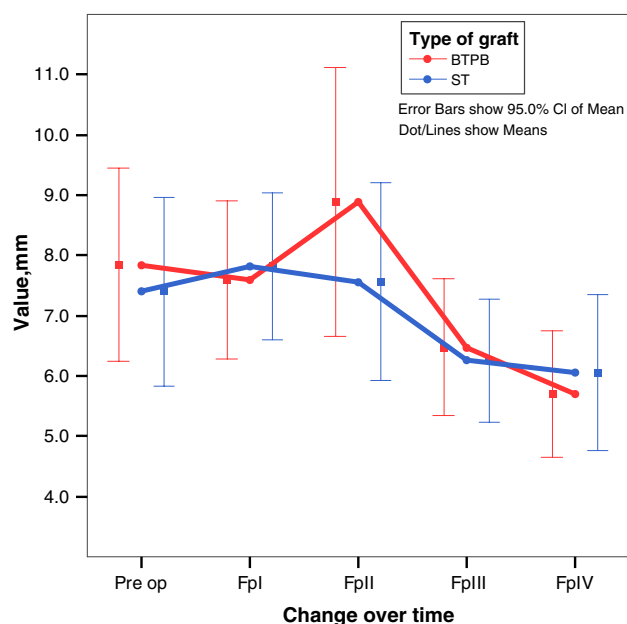


Fig. 3 Absolute values non-involved side ($n = 47$)

Despite many studies of ACL reconstruction using BPTB autografts, few studies have evaluated knee laxity over time. Shelbourne et al. analysed knee laxity over time in 209 patients 2–6 years after ACL reconstruction using BPTB autografts and found no changes in the KT-1000 arthrometer maximum manual testing (MMT) side-to-side difference [37]. Likewise, Salmon et al. found no changes in absolute MMT knee laxity values over time on the injured side between 7 and 13 years after ACL

reconstruction using patellar tendon autografts in their retrospective series involving 33 patients. However, when analysing patients who had undergone a meniscectomy at the time of reconstruction separately, a significant increase in MMT over time was reported [35].

Brandsson et al. reported a slight increase in side-to-side knee laxity over time at 89N between the 2-year and the 4–7-year follow-up using the BPTB autograft. However, at 134N, no such increase was found [4].

To our knowledge, no study analysing knee laxity over time for the HS autograft separately can currently be found in the literature.

In spite of several randomised studies comparing the results after ACL reconstruction using BPTB and HS autografts, only a few have analysed the changes in knee laxity over time.

Gobbi et al. found no major changes in side-to-side laxity between 1 and 3 years in their prospective study involving 80 patients (40 patients in each group) using a computerised laxity instrument applying a force of 200N [15]. In a prospective randomised study, Feller et al. compared BPTB and ST autografts and examined 57 patients at 4 and 8 months and 1–3 years post-operatively. The side-to-side difference was measured at 67 N and 134 N using a KT-1000 arthrometer. Although no statistical analysis was made of laxity over time, no obvious decrease or increase in laxity could be seen [11].

In their 5-year follow-up of 137 patients, Pinczewski et al. found that the manual Lachman's test grade 0, as well as a negative pivot shift test, increased significantly over time during the first three to five post-operative

Table 5 Radiographic assessments classified according to the Ahlbäck and Fairbank rating systems

		Group BPTB $n = 21$	Group ST $n = 23$	Significance
Ahlbäck medial	0	19 (90.5%)	20 (87.0%)	n.s. (0.69)
	1	2 (9.5%)	2 (8.7%)	
	2	0 (0%)	0 (0%)	
	3	0 (0%)	0 (0%)	
	4	0 (0%)	0 (0%)	
	5	0 (0%)	1 (4.3%)	
Ahlbäck lateral	0	19 (90.5%)	22 (95.7%)	n.s. (0.54)
	1	2 (9.5%)	0 (0%)	
	2	0 (0%)	0 (0%)	
	3	0 (0%)	1 (4.3%)	
	4	0 (0%)	0 (0%)	
	5	0 (0%)	0 (0%)	
Fairbank medial flatting	Yes/No	10/11	10/13	n.s. (0.79)
Fairbank medial narrowing	Yes/No	6/15	6/17	n.s. (0.86)
Fairbank medial ridging	Yes/No	9/12	12/11	n.s. (0.54)
Fairbank lateral flatting	Yes/No	3/18	3/20	n.s. (0.91)
Fairbank lateral narrowing	Yes/No	2/19	3/20	n.s. (0.72)
Fairbank lateral ridging	Yes/No	6/15	7/16	n.s. (0.89)

years, in both the BPTB and HS groups [29]. The same group of patients was analysed after 10 years, without any significant changes over time in either group in the Lachman's test, pivot shift test or the mean MMT side-to-side difference compared with the values at the 2-year follow-up. On the other hand, the number of patients with a side-to-side difference of <3 mm in the BPTB group decreased significantly between 2 and 10 years post-operatively, but the corresponding finding was not made in the ST group [30]. In the present study, the decrease in side-to-side difference over time was more pronounced in the BPTB group compared with the ST group, but the difference was not statistically significant (Table 4).

Increased laxity, as a result of graft elongation or graft failure, is regarded as multifactorial and is influenced by mechanical factors such as tunnel positioning, graft impingement, graft fixation and tensioning, graft strength, integrity of secondary stabilisers such as combined ligament injuries or meniscal and cartilage damage. Furthermore, biological and possibly immunological factors can also play a role in changes in knee laxity over time [19, 27]. Changes in laxity, as measured with the KT-1000 arthrometer, not only relate to a process within the graft. Shelbourne et al. suggested that a reduction in ROM, especially a lack of hyperextension, might result in a lower KT-1000 arthrometer measurement [37]. Sernert et al. compared patients with "stiff" knee laxity after ACL reconstruction with patients with normal and increased knee laxity and found that the "stiff" patients had more extension deficit 2–5 years after the reconstruction using BPTB autografts [36]. The development of early degenerative changes and secondary changes in the integrity of ligamentous tissue, as well as the soft tissue envelope, may also influence laxity. The prevalence of radiographically visible degenerative arthritis following an ACL injury is multifactorial and has been reported to range between 40 and 70% in long-term follow-ups [25, 26, 44]. The question of whether the choice of graft may influence the development of arthritic changes is the subject of debate; Pinzowski et al. reported a significantly higher incidence of arthritic changes using BPTB autografts compared with using the HS autograft in their 10-year follow-up of 138 patients evaluated radiographically using the IKDC system [30]. However, in their 7-year follow-up, Lidén et al. reported a comparable prevalence of OA in the BPTB and HS group using the Ahlbäck and Fairbank rating systems in a radiographic evaluation of 113 patients 7 years post-operatively [24]. The prevalence of radiographically visible degenerative arthritis at 7 years in the present study did not differ between the study groups and the impact of degenerative changes on laxity should therefore be similar in both study groups.

The clinical relevance of the present study indicates that knee laxity decreases over time, rather than increasing, irrespective of the type of graft.

The limitation of the present study is the small number of patients, including a follow-up rate of less than 80%, which could represent a type II error.

The strengths are the randomised design and the long and continuous 7-year follow-up.

Conclusion

There were no significant differences in knee laxity measurements between the two study groups pre-operatively or at 7 years. A decrease in knee laxity over time was seen in both the BPTB and HS groups. There was no significant difference between the BPTB group and HS group regarding radiographically visible osteoarthritis at 7 years.

Acknowledgments Financial support was provided by the Research and Development Department of the Northern Älvsborg/Bohus County Council, the Western Sweden County Council research Found and the Swedish Centre for Research in Sports.

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